

Algebra PhD Exam, August 2023

Answer seven problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

- Briefly explain your answers to the following, freely using standard results.
 - Find a polynomial $f(x) \in \mathbb{F}_3[x]$ such that $K = \mathbb{F}_3[x]/(f(x))$ is a field with 27 elements.
 - Show that K is Galois over $\mathbb{F}_3$, and write down the automorphisms of K fixing $\mathbb{F}_3$.
 - For which n is there an extension L/K of fields such that L has 3^n elements?
- Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of degree 5 with exactly three real roots. State and prove a result about the Galois group of the splitting field of $f(x)$ over $\mathbb{Q}$.
- Let $K = \mathbb{Q}(\zeta_{101})$, where ζ_{101} is a primitive 101th root of unity.
 - Show that there is a unique subfield F of K such that $[F : \mathbb{Q}] = 2$.
 - How many subfields of K contain F ? Of these, which are Galois extensions of $\mathbb{Q}$?
- Recall that the Abelianization of a group G is $G^{\text{ab}} := G/[G, G]$.
 - Formulate and prove a universal property for abelianization.
 - Show that abelianization is left adjoint to the forgetful functor from the category of abelian groups to the category of groups.
- Let K be a field and V a K -vector space. Recall that an alternating bilinear map $f : V \times V \rightarrow W$ is a function which is K -linear in each coordinate (i.e. K -multilinear) and such that $f(v, v) = 0$ for all $v \in V$. Show that the functor $\text{Alt}_V^2 : \text{Vec}_K \rightarrow \text{Sets}$ defined by

$$\text{Alt}_V^2(W) = \{f : V \times V \rightarrow W : f \text{ is alternating bilinear}\}$$
 is a representable functor. (The representing object, usually denoted $\Lambda^2(V)$, is an exterior power of V .) Suggestion: A quotient of $V \otimes_K V$.
- Give examples of the following, and briefly explain your answers.
 - a ring R and a short exact sequence of R -modules

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$
 that is not split.
 - a flat $\mathbb{Z}$ -module which is not free.
 - a non-zero injective $\mathbb{Z}$ -module.
- Let R be a ring and M and N be R -modules. Let $P_\bullet$ and $P'_\bullet$ be projective resolutions of M and N . Given a morphism of R -modules $f : M \rightarrow N$, show that there exists a morphism of resolutions $f : P_\bullet \rightarrow P'_\bullet$ extending f . In other words, show there exists $f_0, f_1, f_2, \dots$ making the following diagram commute:

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & P_2 & \xrightarrow{d_2} & P_1 & \xrightarrow{d_1} & P_0 \xrightarrow{\epsilon} M \\
 & & f_2 \downarrow & & f_1 \downarrow & & f_0 \downarrow & f \downarrow \\
 \cdots & \longrightarrow & P'_2 & \xrightarrow{d'_2} & P'_1 & \xrightarrow{d'_1} & P'_0 \xrightarrow{\epsilon'} N
 \end{array}$$

8. What are the dimensions of $\mathbb{Q}[x]/(x^5) \otimes_{\mathbb{Q}} \mathbb{Q}[x]/(x^6)$ and of $\mathbb{Q}[x]/(x^5) \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(x^6)$ as $\mathbb{Q}$ -vector spaces?
9. Let F be a field. Prove that the power series ring

$$F[[x]] = \left\{ \sum_{n=0}^{\infty} a_n x^n : a_n \in F \right\}$$

is a Noetherian ring.

10. Let $B = \mathbb{Q}[x, y]/(xy)$ and $A = \mathbb{Q}[t]$. Note that both have Krull dimension 1. A homomorphism $f : A \rightarrow B$ of rings turns B into an A -module: for $a \in A$ and $b \in B$, $a.b = f(a)b$.
- (a) Explain why the Noether normalization lemma implies that there is a map $A \rightarrow B$ making B into a finitely generated A -module.
 - (b) Show that the map $A \rightarrow B$ sending t to x does not make B into a finitely generated A -module.
 - (c) Find an explicit map $A \rightarrow B$ that makes B into a finitely generated A -module and explain your answer.
11. Let K be a field and consider the function $v : K(x) \rightarrow \mathbb{Z} \cup \{\infty\}$ given by

$$v(f/g) = \deg(g) - \deg(f), \quad v(0) = \infty.$$

- (a) Show that v is a discrete valuation on $K(x)$.
- (b) Show that the valuation ring $\mathcal{O}_v = \{R \in K(x) : v(R) \geq 0\}$ is a local ring. (Do not appeal to general facts about DVR's.)